Modelling Human Driver Responses During an AV-Led Mandatory Merge

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1 Research focus

The experiment contains three vehicles:

- **A:** automated vehicle (AV) in the ending lane;
- **L:** human-driven vehicle in the left target lane;
- **R:** human-driven vehicle behind A.

The thesis focuses on predicting and explaining the responses of L and R. It does not aim to design an AV controller.

The main predictions are:

- whether L yields or continues in lane;
- whether R brakes or maintains speed;
- how the responses develop during the merge;
- whether stress and repeated exposure are associated with different driver profiles.

2 Three proposed models

The study will focus on three model families:

1. **Game-theoretic model:** two pairwise games as baselines, followed by a full three-player game.
2. **Hidden Markov Model:** a sequential model of hidden behavioral phases within a merge.
3. **Latent stress-adaptation model:** an abstract model of behavioral profiles that may change across repeated AV encounters.

The models answer different questions. The game model explains decisions through competing costs. The HMM describes how behavior changes over time within an interaction. The latent profile model describes differences between drivers and possible changes across repeated trials.

3 Model 1: game-theoretic decision model

3.1 What makes it a game

A game contains:

1. players who make decisions;
2. actions available to each player;

3. costs associated with each combination of actions;
4. a rule for predicting which actions will be chosen.

The AV acts first by signalling, changing speed, or beginning the merge. The human drivers observe this behavior and choose their responses. This gives a sequential structure:

$$\text{observed AV action} \longrightarrow \text{human response decision.} \tag{1}$$

If the AV follows one fixed script, its behavior is treated as an observed first move rather than an AV strategy that the thesis must optimize. If the experiment includes several AV strategies, their effects can be modelled explicitly.

3.2 What kind of game this is

The game is non-cooperative. The drivers cannot talk to one another, or to the AV, before the merge, and nothing forces them to keep an agreement. Each player acts to reduce their own cost.

Play is sequential and can run for several stages. The AV moves first. L and R respond, and a response can change the gaps and speeds enough that another decision becomes available. The number of stages is not fixed in advance. It depends on the actions: the interaction continues while a player still has a response to make, and it ends when the merge is resolved. The first estimated models can still score one response at a time. The stage structure is what links those responses.

Information is incomplete for every player, not only for one of them. A driver does not know another road user's cost weights. Each driver instead holds a prior belief, and those priors can differ. A belief can be revised as the stages proceed. Two revision rules should be kept apart, because they answer different questions:

- Bayesian updating revises the prior belief about what kind of driver the other road user is. Within one merge there is little repeated evidence about a stable type, so this update is expected to be infrequent.
- Behavioral updating revises what the driver expects the other to do next, after a move has been seen, during the current merge. This is the more feasible rule for these data. How that revision should be calculated is still an open question.

A round effect sits outside a single merge. The first encounter with the AV may change the decision in the next encounter, for example through the exposure term already allowed in α_i, or through the latent profile in Section 5. That is an effect across trials, not an update inside one interaction.

The same interaction can be read in three ways, and they should be compared rather than combined without comment:

1. a perfect-information game, in which previous moves are observed and the costs are treated as known;
2. a Bayesian game, in which the cost weights are hidden and a prior belief stands in for them;
3. a behavioral game, in which the driver follows the probabilistic choice rule of Section 3.6 rather than the lowest cost on every occasion.

A sequential game that combines hidden costs with beliefs updated along the path is a perfect Bayesian game. Whether the recordings support that solution, or only the behavioral rule with observed moves, is a comparison rather than an assumption.

3.3 Two two-player baseline games

The first baseline separates the three-vehicle situation into two simpler games.

3.3.1 Game A–L

Players:

- A, which performs the mandatory merge;
- L, which responds from the target lane.

Initial human actions:

$$a_L \in \{\text{Yield}, \text{Continue}\}. \tag{2}$$

This game predicts whether L creates space for A or maintains its current position and speed.

3.3.2 Game A–R

Players:

- A, which may signal, slow down, and move laterally;
- R, which follows A in the ending lane.

Initial human actions:

$$a_R \in \{\text{Brake}, \text{Maintain}\}. \tag{3}$$

This game predicts whether R opens the following gap or maintains its current speed.

These pairwise games are useful baselines because they are easier to estimate. They test whether each human response can be explained using only the human and the AV.

3.4 Example cost structure

For player i and candidate action a, an example cost is:

$$C_i(a) = w_{s,i}S_i(a) + w_{e,i}E_i(a) + w_{c,i}K_i(a) + \underbrace{w_{I,i}I_i(a)}_{\text{optional}}. \tag{4}$$

The components are:

- $S_i(a)$: safety cost;
- $E_i(a)$: efficiency cost, such as speed or time lost;
- $K_i(a)$: comfort cost, such as strong acceleration or braking;
- $I_i(a)$: optional interaction cost, such as leaving an insufficient merge gap;
- $w_{s,i}, w_{e,i}, w_{c,i}, w_{I,i}$: weights describing the importance of each component.

A lower value means that the action is less costly and therefore more attractive.

The formulas below are only examples. They show how the model could be constructed, but they are not final definitions. They can be replaced after the dataset, vehicle geometry, and signal quality have been examined.

3.4.1 Example safety cost

One possible safety cost is inverse time to collision:

$$S_i(a) = \frac{1}{\max(\text{TTC}_i(a), \varepsilon)}. \tag{5}$$

Small TTC produces a large safety cost. If TTC is not suitable for a specific vehicle arrangement, it can be replaced by projected minimum gap, time headway, or required braking.

3.4.2 Example efficiency cost

One possible efficiency cost is the squared loss of speed:

$$E_i(a) = \left(v_i^{\text{desired}} - v_i^{\text{after}}(a)\right)^2. \tag{6}$$

This expresses that braking or yielding may reduce efficiency. Other possible definitions include delay, time spent below desired speed, or distance lost.

3.4.3 Example comfort cost

One possible comfort cost is the magnitude of the required acceleration:

$$K_i(a) = |a_i(a)|. \tag{7}$$

Strong acceleration or braking gives a larger comfort cost. This formula can later be replaced by a measure that better matches the available data.

3.4.4 Optional interaction cost

The interaction term can be included only if it adds a clear and measurable concept that is not already captured by safety and efficiency. For example:

$$I_L(a) = \begin{cases} 0, & \text{if L creates a sufficient merge gap,} \\ 1, & \text{if L leaves an insufficient merge gap.} \end{cases} \tag{8}$$

This is an example definition. The term should be removed if it duplicates the safety cost or cannot be measured reliably. It should not be interpreted as a direct measurement of kindness or intention.

3.5 Decision from the cost

The simplest model predicts that a player selects the lowest-cost action:

$$a_i^* = \operatorname*{argmin}_a C_i(a). \tag{9}$$

For example:

$$\text{predict Yield if } C_L(\text{Yield}) < C_L(\text{Continue}). \tag{10}$$

This deterministic rule is easy to interpret, but human decisions are not perfectly consistent.

3.6 Probabilistic human choice

A more realistic model converts costs into action probabilities:

$$P_i(a) = \frac{\exp[-\alpha_i C_i(a)]}{\sum_b \exp[-\alpha_i C_i(b)]}. \tag{11}$$

The parameter α_i describes how consistently the player chooses the lowest-cost action:

- small α_i: choices are more variable;
- large α_i: the lowest-cost action is selected more consistently.

Human factors may influence α_i. A simple conceptual form is:

$$\alpha_i = f(\text{stress}, \text{AV exposure}, \text{reaction-time history}). \tag{12}$$

For example, stress may make choices less consistent, while repeated exposure may make the AV more predictable. The direction should be estimated from the data rather than assumed.

Only information available before the predicted decision should be used. Therefore, "reaction-time history" means reaction time from earlier trials. If the model is updated during the current trial, elapsed time up to the present moment may also be used. The final reaction time of the current decision must not be used to predict that same decision.

Human factors could alternatively influence the cost weights. For example, a more cautious driver profile may assign a larger weight to safety.

3.7 Full three-player game

The main game-theoretic model will attempt to represent A, L, and R together:

$$\mathcal{G} = \{A, L, R\}. \tag{13}$$

Each player has a cost that depends on the full combination of actions:

$$C_A = C_A(a_A, a_L, a_R), \tag{14}$$
$$C_L = C_L(a_L, a_A, a_R), \tag{15}$$
$$C_R = C_R(a_R, a_A, a_L). \tag{16}$$

This model can represent links that the pairwise games omit. For example:

1. L opens or closes the target-lane gap;
2. this affects how A performs the merge;
3. A's resulting speed change affects the following risk for R.

The full model will be compared with the two pairwise baselines. It is useful only if it improves prediction and if the data show that the three vehicles influence one another. If the AV has only one fixed behavior and does not adapt, the model should be described as a joint human-response model rather than a complete three-player equilibrium.

4 Model 2: Hidden Markov Model

4.1 Purpose

The game model predicts which action has the lowest expected cost. An HMM addresses a different question: which hidden behavioral phase is the interaction currently in?

Possible hidden phases are:

$$\text{Approach} \longrightarrow \text{Anticipation} \longrightarrow \text{Yield, Continue, or Conflict} \longrightarrow \text{Resolution}. \tag{17}$$

The phases are hidden because they are not labelled directly in the recorded data. They are inferred from observations such as:

- speed and acceleration;
- gaps and relative speed;
- AV indicator and lateral movement;
- human Yield, Continue, Brake, or Maintain actions;
- stress, if synchronized reliably.

4.2 How it works

The model assumes that the current hidden phase mainly depends on the previous phase:

$$P(z_t \mid z_{t-1}). \tag{18}$$

Each phase has a characteristic pattern of observable data:

$$P(\text{observations at } t \mid z_t). \tag{19}$$

The HMM estimates:

- the probability of moving from one phase to another;
- the observations normally associated with each phase;
- the most likely phase sequence for each merge.

4.3 Why it may be useful

The HMM can:

- divide continuous trajectories into recurring phases;
- reveal common action sequences;
- compare how quickly different drivers move from anticipation to action;
- provide a sequential baseline that does not assume strategic cost minimization.

Its main limitation is that it describes temporal patterns but does not by itself explain why one action is strategically preferable. Hidden states are statistical patterns, not direct measurements of intention.

5 Model 3: latent stress-adaptation model

5.1 Purpose

This model focuses on more abstract human factors. It asks whether stress, repeated exposure, and previous interaction outcomes are associated with different behavioral profiles.

At exposure e, participant p has an unobserved profile:

$$z_{p,e} \in \{\text{Profile 1}, \text{Profile 2}, \text{Profile 3}\}. \tag{20}$$

The profiles should initially have neutral names. After fitting, the observed behavior may support descriptions such as:

- cautious or stressed;
- balanced or adaptive;
- assertive or habituated.

These labels describe behavioral patterns. They are not evidence of fixed personality traits.

5.2 Changing profile across exposures

The probability of the current profile can depend on:

$$\begin{aligned} P(\text{current profile}) = f(&\text{previous profile},\ \text{stress},\ \text{exposure},\\ &\text{previous outcome},\ \text{previous reaction time}). \end{aligned} \tag{21}$$

This allows a participant to move between profiles. For example, repeated predictable encounters may be associated with lower stress and a more confident response pattern. A difficult previous encounter may instead be associated with a more cautious next response.

These are hypotheses to test. The model must not assume in advance that exposure always reduces stress or makes drivers more assertive.

5.3 Abstract baseline and connection to the game

The latent stress-adaptation model can first be used as an abstract baseline using:

- pre-decision stress;
- exposure number;
- previous outcome;
- previous human action;
- previous reaction time;
- gaze or attention, if reliable.

It deliberately excludes physical variables such as TTC and gap. This tests how much can be predicted from human factors and interaction history alone.

The latent profile can then be connected to the game model. Different profiles may have different cost weights or different α values:

$$\text{profile} \longrightarrow \{w_s, w_e, w_c, w_I, \alpha\} \longrightarrow \text{action probability}. \tag{22}$$

For example, a cautious profile may give greater weight to safety, while an efficiency-oriented profile may give greater weight to maintaining speed.

5.4 Why it may be useful

This model can:

- separate temporary stress from more stable differences between participants;
- test whether profiles change across repeated AV encounters;
- provide a human-factor baseline for the physical game model;
- help explain why drivers facing similar traffic conditions choose different actions.

Its main limitations are:

- stress must be measured before the decision if it is used as a predictor;
- enough repeated trials are required to estimate profile changes;
- profile names must follow from the results;
- the model cannot prove that a participant's real personality changed.

6 How the three models relate

The proposed analysis has a clear progression:

1. Fit the two pairwise games A–L and A–R as understandable baselines.
2. Fit the full A–L–R game and test whether joint interaction improves human-response prediction.
3. Fit an HMM to identify recurring behavioral phases within each merge.
4. Fit the latent stress-adaptation model across repeated trials.

5. Test whether latent profiles improve the game model by changing its weights or choice-consistency parameter.

The three models are complementary:

- game theory explains choices through costs and strategic dependence;
- the HMM explains the temporal sequence within an interaction;
- the latent stress-adaptation model explains abstract driver differences and changes across exposures.

The full three-player game is the intended main model. The two pairwise games provide a simpler baseline. The HMM and latent stress-adaptation model provide alternative and complementary descriptions of human behavior.